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Abstract

Hemolytic uremic syndrome (HUS) and thrombotic thrombocytopenia purpura (TTP) are rare diseases sharing a common pathological feature, thrombotic microangiopathy (TMA). TMA is characterized by microvascular thrombosis with consequent thrombocytopenia, microangiopathic hemolytic anemia and/or multiorgan dysfunction. In the past, the distinction between HUS and TTP was predominantly based on clinical grounds. However, clinical presentation of the two syndromes often overlaps and, the differential diagnosis is broad. Identification of underlying pathogenic mechanisms has enabled the classification of these syndromes on a molecular basis: typical HUS caused by Shiga toxin-producing *Escherichia coli* (STEC-HUS); atypical HUS or complement-mediated TMA (aHUS/CM-TMA) associated with genetic or acquired defects leading to dysregulation of the alternative pathway (AP) of complement; and TTP that results from a severe deficiency of the von Willebrand Factor (VWF)-cleaving protease, ADAMTS13.

The etiology of TMA differs between pediatric and adult patients. Childhood TMA is chiefly caused by STEC-HUS, followed by CM-TMA and pneumococcal HUS (Sp-HUS). Rare conditions such as congenital TTP (cTTP), vitamin B12 metabolism defects, and coagulation disorders (diacylglycerol epsilon mutation) present as TMA chiefly in children under 2 years of age. In contrast secondary causes and acquired ADAMTS13 deficiency are more common in adults. In adults, compared to children, diagnostic delays are more frequent due to the wide range of differential diagnoses. In this review we focus on the three major forms of TMA, STEC-HUS, aHUS and TTP, outlining the clinical presentation, diagnosis and management of the affected patients, to help highlight the salient features and the differences between adult and pediatric patients which are relevant for management.

Semin Nephrol 43:151436 © 2023 Elsevier Inc. All rights reserved.

Key words: Hemolytic uremic syndrome, thrombotic thrombocytopenic purpura, complement

INTRODUCTION

Thrombotic thrombocytopenic purpura (TTP) and hemolytic uremic syndrome (HUS) are rare diseases sharing a common pathological feature, the thrombotic microangiopathy (TMA), which is characterized by thickening of the microvasculature wall, with intraluminal platelet thrombosis and obstruction of the vessel lumina, leading to hypoperfusion and ischemia of the affected organ(s), especially the kidney and brain (Fig. 1). Laboratory features are thrombocytopenia and microangiopathic hemolytic anemia, caused by platelet consumption in the microcirculation, and erythrocyte fragmentation in the occluded vessels.¹

The clinical presentation of TMA varies widely. The conventional distinction between TTP and HUS, based on predominant neurologic involvement and more severe thrombocytopenia in TTP, and more severe acute kidney

injury (AKI) in HUS, is seldom clear-cut, and the clinical presentations can overlap.^{2,3} The identification of the pathogenic mechanisms has enabled the classification of these syndromes on a molecular basis, directing clinicians to specific treatments. HUS is most commonly triggered by *Escherichia coli* strains that produce Shiga-like toxins (STEC-HUS, Fig. 2).⁴ Streptococcus is the most common infection that causes non-diarrheal HUS in children (Sp-HUS, Fig. 2). Genetic or acquired abnormalities that lead to dysregulation of the alternative pathway (AP) of complement are associated with atypical HUS (aHUS, Fig. 2).^{5,6} TTP is caused by severe deficiency of ADAMTS13, a plasma metalloprotease that cleaves von Willebrand Factor (VWF) multimers.⁷ TTP is most often immune-mediated (iTTP), due to inhibitory antibodies against ADAMTS13, and rarely, congenital (cTTP), due to ADAMTS13 pathogenic variants (Fig. 2).

Some forms of TMA, especially in adults, are associated with coexisting diseases or conditions, such as malignant hypertension, autoimmune diseases, malignancy, drugs, infections and organ transplantation, and are termed secondary TMA (sTMA, Fig. 2). The distinction between the causes and the precipitating factors is often not absolute.⁸ Indeed, many of the above conditions may either directly cause TMA or trigger it in genetically predisposed individuals (e.g., pregnancy-associated or post-transplantation TMA).

Independent of the underlying pathogenesis, endothelial cell damage is critical in the sequence of events that

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0270-9295/ - see front matter

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<https://doi.org/10.1016/j.semnephrol.2023.151436>

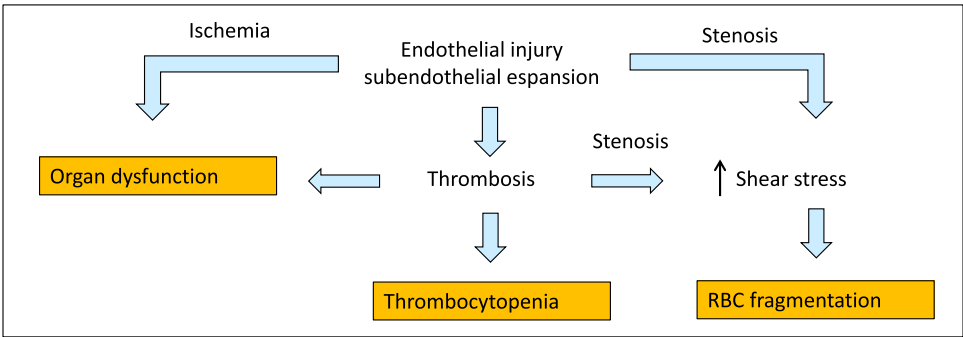


Figure 1. From endothelial injury to clinical manifestation of thrombotic microangiopathy (TMA).

lead to TMA. Loss of endothelial thromboresistance, leukocyte adhesion to injured endothelium, altered VWF cleavage, complement activation, and increased vascular shear stress then sustain and amplify the microangiopathic process.

INFECTION-ASSOCIATED HUS

STEC-HUS

Epidemiology and etiology

The incidence of STEC-HUS varies from 0.5 cases/100,000 population in Finland to

12 cases/100,000 population in Argentina. STEC-HUS is chiefly seen in young children (Table 1).^{9–11}

E.coli 0157:H7 was the most common cause of worldwide outbreaks of STEC-HUS, accounting for 70% of cases up to the mid-2000s. Since 2010, the epidemiology of STEC-HUS has changed, with over 100 non-0157 pathotype strains reported in worldwide outbreaks.^{12,13} In 2011, Europe, particularly Germany, experienced one of the largest STEC-HUS outbreaks, traced to the aggressive O104:H4 *E.coli* strain.¹⁴ Almost 90% of infected subjects were adults, and 22% of them developed HUS. After 2011, the main outbreaks of STEC-HUS involving adults were ascribed to non-O157 strains, with the predominance of serogroups O26, O103, O146, O91, O111, O145, O55, and O12.^{9,15,16} In Europe, non-

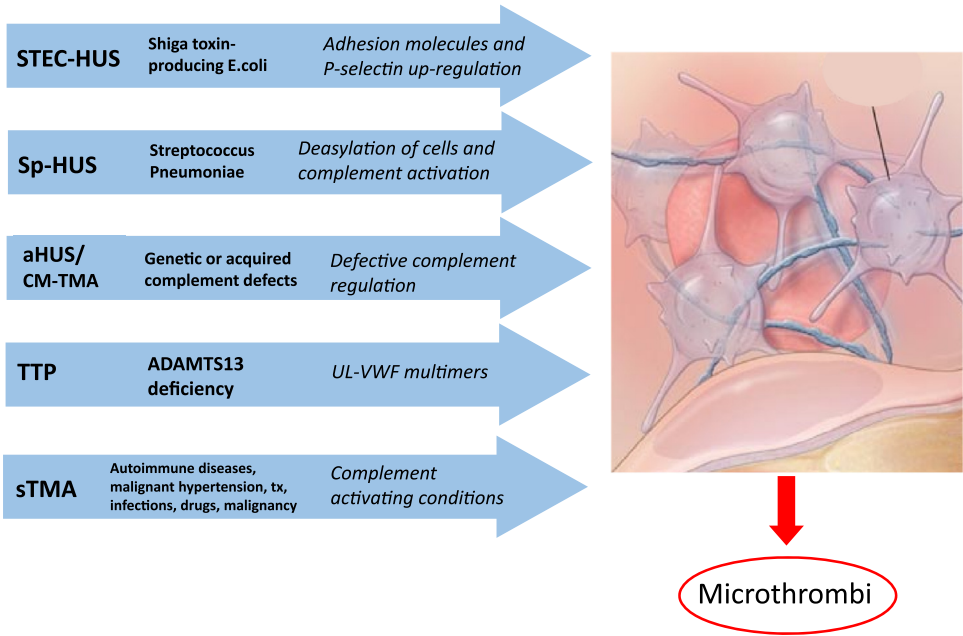


Figure 2. Thrombotic microangiopathy classification. Based on underlying pathogenic mechanisms, the thrombotic microangiopathies, were classified as: Shiga toxin (Stx) producing *E. coli* associated HUS (STEC-HUS), HUS associated with *Streptococcus pneumoniae* infection (Sp-HUS), atypical HUS or complement-mediated TMA (aHUS or CM/TMA), associated with genetic or acquired dysregulation of the complement alternative pathway; thrombotic thrombocytopenic purpura (TTP) resulting from deficiency or inhibition of a disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13 (ADAMTS13), and secondary forms of TMA (sTMA). All forms share the common features of endothelial injury and platelet thrombosis in the microcirculation of several organs. UL–vWF ultra-large multimers of von Willebrand factor.

Table 1. Epidemiology of different forms of TMA

TMA Spectrum	Incidence (Cases/year)	Age of onset	References
Infection-associated HUS			
STEC-HUS	0.5 per 100,000 in Finland 12 per 100,000 in Argentina	60% <16 yrs 26% 16-59 yrs 20% > 60 yrs	9–11
Sp-HUS	0.015 to 0.065 per 100,000 (5-10% of all cases of HUS)	Childhood (typically < 2 yrs)	57
aHUS			
Complement-mediated HUS	0.5 to 2 per million (0.26 to 0.75 per million < 20 yrs)	41-58% in adulthood	76–78
non complement-mediated HUS	0.009 per million (associated with DGKE variants)	typically < 2 yrs	116,120
Metabolism-associated HUS	very rare (<5% of all cases of TMA)	neonatal period or infancy	122–124
TTP			
immune-mediated TTP	~ 2 to 6 per million (95% of all TTP cases)	95% in adulthood (average age 40 yrs)	135,136
congenital TTP	~ 0.5 to 2 per million (5% of all TTP cases)	mainly in childhood (<10 yrs) and obstetrical patients	135–137

O157 serotypes have become more common than O157:H7 even in children.¹⁷

Pathogenesis

After exposure to STEC, 3-9% (in sporadic infections) and 20% (in epidemic forms) of individuals develop overt HUS. STEC produce two major types of Shiga-like toxins (Stx): Stx1 and Stx2, each with different subtypes (Stx1a-1d and Stx2a-2g). Stx2, especially Stx2a, is the most frequently responsible for severe human disease. STEC colonize the intestinal mucosa and release Stx, which translocates through the epithelium into the circulation.^{11,18} Blood cells or blood-derived microvesicles serve as Stx-carriers from the intestine to target organs (chiefly the kidneys, but also brain and lungs) that express the Stx-specific receptor, Gb3. Once inside cells, Stx inhibits ribosomal function and protein synthesis. The toxin also upregulates the expression of proinflammatory proteins involved in both the recruitment and adhesion of leukocytes and thrombus formation at the site of endothelial injury.¹⁸ In addition, Stx induces the release of VWF from endothelial cells, and inhibits the cleavage of VWF by ADAMTS13,¹⁹ favoring platelet adhesion and clot formation in the microvasculature. Emerging evidence suggests that complement activation may contribute to the pathophysiology of STEC-HUS. *In vitro* studies have shown that Stx2 directly activates the AP at endothelial cell surface.¹⁸ All these events cause a shift of endothelial cells toward a prothrombotic phenotype.

Clinical presentation

STEC-HUS is typically characterized by prodromal diarrhea followed by AKI, in children. The average interval between *E. coli* exposure and illness is 4 days (range: 1-10 days). Symptoms usually begin with abdominal cramps and watery diarrhea, which may become

hemorrhagic in 60-70% of cases within 1-5 days.²⁰ Progression to HUS usually occurs 7-10 days after the onset of symptoms.

Bloody diarrhea is not a distinctive feature of STEC-HUS in adults, indeed it does not occur in 30%-40% of patients.^{9,21–23} Non-bloody diarrhea and vomiting are more common (92% of cases).²⁴

AKI in STEC-HUS patients ranges from asymptomatic urine sediment abnormalities to severe AKI and end-stage kidney failure (ESKF). Approximately 30-60% of children with STEC-HUS require renal replacement therapy (RRT). RRT was required in 43% of adult patients from the Oklahoma registry²⁵ and in 63.5% of the French cohort.²⁶

Ten to 25% of children with STEC-HUS show neurological involvement, in the form of irritability, lethargy, encephalopathy, seizures, stroke and focal deficits.²⁷ While most authors report that these manifestations are associated with severe AKI, long-term neurological sequelae, and adverse outcomes, one recent study on 202 children found satisfactory outcomes in patients with severe neurological involvement.^{27,28} Severe neurological manifestations are prevalent in adults (Table 2). About 50% of the patients with STEC-HUS from both the O104:H4 German outbreak²⁹ and the adult French cohort²⁶ and 67% of the adult patients from the Oklahoma registry,²⁵ exhibited neurologic symptoms such as cognitive impairment, seizures and coma. In elderly patients, severe neurological complications represent a prognostic clinical marker of unfavorable outcome²¹ (Table 2).

More than half of pediatric and adult patients present with elevated blood pressure. In children cardiovascular manifestations include elevated troponin, myocardial dysfunction, and pericardial effusion.²⁸

Acute gastrointestinal complications in children include hemorrhagic colitis, bowel ischemia/necrosis, and perforation. TMA may cause pancreatitis, pancreatic

Table 2. Comparison of clinical features at disease presentation between adults and children in STEC-HUS, aHUS and TTP

		STEC-HUS	aHUS	TTP
Neurological Symptoms	Adults	50%-67%	8%-25%	>80%
	Children	10%-25%	16%-24%	
Gastrointestinal symptoms	Adults	>90%	33%	35%
	Children	>90%	47%	
Cardiovascular complications	Adults	<10%	3%-10%	25%
	Children	<10%	3%-10%	
Dialysis	Adults	43%-65%	81%	16%
	Children	30%-60%	59%	

necrosis, and beta cell dysfunction, leading to acute or chronic insulin-dependent diabetes. Other extrarenal manifestations include elevated transaminases and/or bilirubin, and rhabdomyolysis.²⁸

Investigations and Diagnosis

Early diagnosis of STEC-HUS may be difficult, especially in adults, due to the overlap of some clinical and biological features between STEC-HUS, aHUS and TTP. Therefore, when infection by STEC is suspected, STEC screening is recommended.

Diagnosis depends on the detection of *E.coli* O157 and other STECs and their products in stool culture²⁰ (within 6-days following the onset of symptoms). Real-time PCR has improved the identification of non-O157:H7 strains, and can detect Stx-encoding genes directly on stools specimens following enrichment, up to 20 days after onset of the symptoms. It is considered the most sensitive and specific investigation to confirm diagnosis.^{30,31}

Positive serology for antibodies to O157 or other specific strain-derived lipopolysaccharide supports a diagnosis of STEC-HUS. This can be useful when stool for STEC is negative, due to late collection. However, false-positives may arise due to antibodies pre-formed during antecedent STEC exposure.²⁰

Neutrophilic leukocytosis and elevated levels of C-reactive protein and d-dimer are frequent in patients with STEC-HUS, mimicking sepsis. A transient decline in C3 levels may be seen in severe STEC-HUS.³²

Retrospective data on 100 children with Stx2-positive bloody diarrhea suggested that urine dipstick monitoring for hemoglobinuria/hematuria enables detection of STEC-HUS.³³ Hemoconcentration, was associated with subsequent complicated STEC-HUS (major neurological symptoms, need for dialysis).³⁴

Management

Management for STEC-HUS is largely supportive and nonspecific, relying on the management of anemia, AKI, and electrolyte and water imbalance. Patients with severe

HUS must be monitored carefully, including urine output, weight, volume status, cardiovascular/respiratory function, and signs of central nervous system or other organ involvement.

Almost 80% of pediatric and adult patients require red blood cell transfusion, although children appear to require less in total.^{9,35} To reduce transfusions, treatment with recombinant erythropoietin has been advocated, although its favorable effect is still controversial.³⁶ Restricted use of platelet transfusion is recommended.^{37,38} Heparin and anti-thrombotic agents may increase the risk of bleeding and should be avoided.

ACEi may be beneficial in the long-term for adult or pediatric patients who develop chronic kidney disease (CKD) or proteinuria after an episode of STEC-HUS, but may not be the treatment of choice for hypertension in the acute phase given the change in glomerular hemodynamics and risk of hyperkalemia.^{9,39}

Specific treatments

Therapeutic plasma exchange (TPE) has been used during outbreaks with conflicting results.⁴⁰ TPE appeared to ameliorate the course of the disease in patients with severe AKI and neurological involvement.⁴¹ However, most reports of early use of TPE, suggesting improved outcomes in children and lower case fatality rates in the elderly with STEC-HUS, were of low quality with risk of observer bias.⁴²

The use of antibiotics is controversial. In adult patients from the German O104:H4 epidemic, the use of azithromycin, a protein and cell wall synthesis inhibitor, was associated with a shortened STEC carriage.⁴³ In contrast, several bactericidal antibiotics (quinolones, trimethoprim, and furazolidone), as potent inducers of the expression of Stx genes, lead to more Stx release increasing the likelihood and severity of STEC-HUS.^{44,45}

Available studies do not support a role for any secondary prevention strategy, namely antibiotics, bovine milk, Shiga toxin inhibitor, or monoclonal neutralizing antibodies, in preventing HUS in patients with STEC diarrhea.⁴⁶

Evidence that uncontrolled complement activation may contribute to microangiopathic lesions of STEC-HUS led to attempts of complement inhibitor therapy with the anti-C5 monoclonal antibody, eculizumab. However, the effectiveness of eculizumab in STEC-HUS remains unclear. In two large series from the O104:H4 adult outbreak in Germany, eculizumab therapy with TPE did not improve outcomes compared to TPE alone.⁴⁷ However, these data are retrospective and patients given eculizumab were the most severely ill. On the other hand, early treatment with eculizumab in 9 adult patients in the O104:H4 outbreak in France was associated with a rapid recovery.⁴⁸ Evidence from 16 case reports appears to support the use of eculizumab in patients with severe STEC-HUS and neurological or multiorgan dysfunction.⁴⁹ Prospective controlled trials in France (ECULISHU, NCT02205541) and in England (ECUSTEC, ISRCTN89553116) have recently concluded and the results will help clarifying whether eculizumab is beneficial in treating severe forms of STEC-HUS.

Long-term outcomes

Meta-analysis of 49 studies conducted before 2003, including 3476 children with STEC-HUS followed for an average of 4.4 years, indicated incidence of death and CKD or proteinuria in 12% and 25% of patients, respectively. However, with improved access to and timely initiation of RRT and supportive care, the acute mortality in children with STEC-HUS has declined to 1% to 4% recently.⁵⁰ Mortality is secondary to severe neurological injury, and less often to pulmonary edema, cardiac failure, hyperkalemia or bowel necrosis. Between 20% and 50% of children with STEC-HUS have long-term sequelae, chiefly in the form of mild CKD. Approximately 3% develop ESKF, while 20-40% show persistent proteinuria and 5-15% have persistent hypertension.⁵⁰⁻⁵² Neurological involvement, dehydration, high hematocrit and leukocytosis have been linked to adverse long-term outcomes.⁵³ Proteinuria at 1-yr after the acute illness was associated with CKD.⁵² Children developing stroke or severe anatomical brain injury during acute STEC-HUS usually have significant long-term neurological damage. At 3-6 months following recovery from O104:H4-associated STEC-HUS, changes in overall performance, abnormalities of cognition/behavior and reduced performance and physical strength were reported in 55%, 67% and 50% of children, respectively.⁵⁴

Meta-analysis of the main 11 outbreaks of STEC-HUS between 1984 and 2015 reported death in 9% of adults,²¹ which was mainly caused by cerebral hemorrhage, ischemic stroke, cardiac involvement and encephalopathy, along with other complications such as sepsis. About 50% of adult cases of STEC-HUS fully recover from the acute illness: those who do not fully recover may have long-term renal and neurological sequelae. After the O104:H4 outbreak, proteinuria,

hypertension and reduced estimated glomerular filtration rate (eGFR) <90 ml/min were observed at 1-year follow-up in 27%, 25% and 47% of adult patients, respectively.⁵⁵ Long-term neuropsychological symptoms were observed in 70% of adults after the O104 outbreak, an unexpectedly high frequency.⁵⁶ Sixty-one percent of the patients scored borderline or low on neuropsychological tests and 25% exhibited cognitive decline.^{2,56}

Pneumococcal HUS

HUS associated with *Streptococcus pneumoniae* infections (Sp-HUS) comprises 5-10% of all cases of HUS.⁵⁷ The complication is observed in ~0.5% of invasive pneumococcal disease, typically affecting children less than two years-old with pneumonia/empyema or meningitis.⁵⁸ While the incidence of Sp-HUS (Table 1) has declined substantially in developed countries with routine pneumococcal vaccination, it remains an important cause of HUS in developing regions with poor immunization coverage, and due to emergence of non-vaccine/replacement serotypes.^{57,59}

Pathophysiology

Sp-HUS manifest as result of Pneumococcal neuraminidase deasialylation of glycoproteins expressed on erythrocytes, platelets, glomerular endothelial cells and kidney epithelial cells, exposing the Thomsen-Friedenreich antigen (T-antigen).⁵⁸ T-antigen is recognised by preformed IgM antibodies, thus initiating events that culminate in Sp-HUS. This antigen-antibody reaction occurs more frequently in children, and causes polyagglutination of red blood cells *in vitro*, resulting in a positive Coomb's test. Complement activation due to complement factor H (FH) dysfunction,^{60,61} decreased sialylation of transferrin and IgA1 O-glycans, and increased expression of galectin-3, might play roles in pathogenesis.^{62,63}

Clinical features

The triad of HUS appears 3-13 days after streptococcal infection.⁶³ The condition is often misdiagnosed as pneumococcal sepsis or disseminated intravascular coagulation (DIC), since affected children may have bleeding manifestations and prolonged prothrombin and activated partial thromboplastin time. Compared to patients with STEC-HUS, children with Sp-HUS show more severe and prolonged thrombocytopenia and AKI; extrarenal complications including hepatic dysfunction, are more common.^{58,63}

Diagnosis

The diagnosis of Sp-HUS is challenging. The diagnostic yield of tests for invasive pneumococcal disease is low;

e.g., blood culture is positive in only 10-30% cases.⁶⁴ A positive direct Coombs test, seen in 60-90% cases, suggests Sp-HUS, but is non-specific. The activation of T-antigen is evaluated by the fluorescence-labeled peanut lectin agglutination method.^{65,66}

Spinale et al. provided case definitions for the recognition of Sp-HUS in critically ill patients.⁵⁸ A *definite* case of Sp-HUS is one with evidence for HUS along with that for invasive *S. pneumoniae* infection, in the absence of features of DIC. Patients fulfilling the former criteria but with evidence of DIC are categorised as *probable* cases if there is evidence of T-activation. A patient with pneumonia, meningitis, or other evidence of invasive infection without bacteriological confirmation is considered as possible Sp-HUS if there is evidence for HUS, along with either T-activation with or without DIC.

Management

Supportive therapy is the mainstay of management.⁶⁷ Early initiation of antibiotic therapy decreases mortality.

The combination of third-generation cephalosporins and vancomycin is initiated in severe cases, and modified later based on results of culture and sensitivity tests. If transfusion of red blood cells is essential, these should be washed to remove preformed IgM antibodies against T-antigen.⁶⁸ The role for TPE is controversial; while apheresis would remove neuraminidase, fresh frozen plasma is best avoided for it might contain preformed IgM antibodies that could aggravate the microangiopathy. Eculizumab therapy data are limited and conflicting.⁶⁷ Widespread immunization coverage and surveillance for disease by replacement serotypes might be useful preventive strategies.⁵⁷

HUS associated with other infections

Infections with viruses, bacteria, fungi and parasites in association with HUS have been widely reported, including H1N1 influenza, HIV, cytomegalovirus, Epstein-Barr virus, varicella zoster virus, parvovirus B19, dengue fever, falciparum and vivax malaria, and most recently,

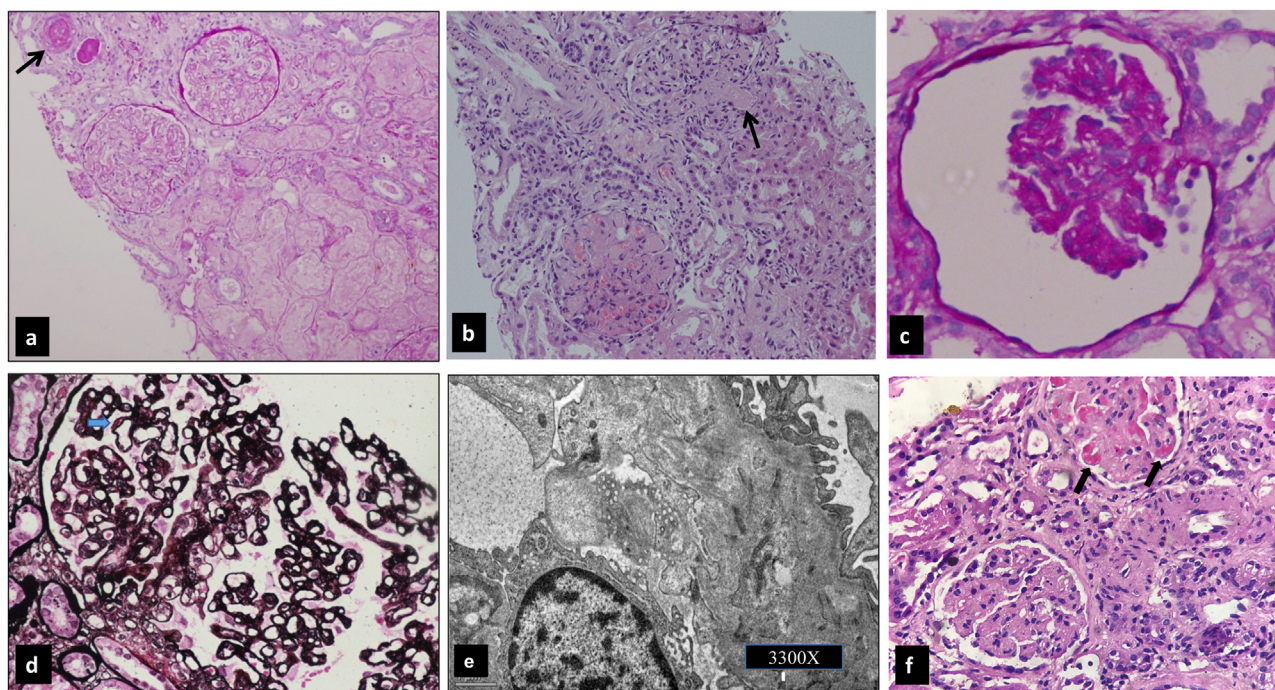


Figure 3. Spectrum of thrombotic microangiopathy (TMA) in children with hemolytic uremic syndrome (HUS): (a) Patchy acute cortical necrosis with necrosed glomeruli and adjacent tubules, with fibrin within lumina of glomerular capillaries and an arteriole at the border zone of the necrotic and viable areas showing an intraluminal fibrin thrombus occluding the lumen (arrow), in a 6-year old girl with TMA associated with *Plasmodium vivax* malaria (Periodic acid Schiff, x 100 original magnification). (b) Extensive mesangiolytic associated with fibrin deposits within glomerular capillary lumina and occlusion of the feeding arteriole (arrow) in a 11-year old boy with atypical HUS secondary to antibodies to complement factor H (Hematoxylin and eosin, x 150 original magnification). (c) Glomerulus showing tuft shrinkage in chronic thrombotic microangiopathy due to homozygous *CFH* pathogenic variant (Periodic acid Schiff, x 250 original magnification); (d) Glomerulus showing extensive capillary wall reduplication (arrow) (Jones's silver methenamine 400x magnification) and (e) capillary wall showing mesangial cytoplasmic interposition in the subendothelial region, without any electron-dense deposits (Uranyl acetate-lead citrate 3300x) in a 10-yr old boy with DGKE nephropathy secondary to a homozygous pathogenic *DGKE* variant. (f) Glomeruli with mesangiolytic and capillary lumina occluded with fibrin in a 15-yr-old boy with post-transplant TMA and no defects in complement or cobalamin pathway (Hematoxylin and eosin, x 100 original magnification) [Images courtesy Dr Geetika Singh, Department of Pathology, All India Institute of Medical Sciences, New Delhi]

SARS-CoV2.^{69–71} The pathogenesis is complex and comprises a direct effect on endothelial cells and complement activation [Fig. 3(a)].

Attributing HUS to systemic infections is tricky, particularly since bacterial sepsis and tropical infections, including malaria, leptospirosis and dengue, often mimic HUS. In the Oklahoma TTP-HUS registry, 7% of patients diagnosed during 1989–2010 were later reclassified as systemic infections.⁷² It seems prudent to diagnose TMA if features of microangiopathic hemolysis are present without concomitant fever or evidence of acute infection.⁷³ Since infectious prodromes are well recognized triggers of TMA in patients with complement dysregulation, evaluation for underlying genetic defects is necessary. Therapy is usually focused on treatment of infection, although benefit with TPE and/or eculizumab is reported.^{74,75}

Atypical Hemolytic Uremic Syndrome

Epidemiology

The annual incidence of aHUS ranges from 0.5 to 2 per million population (Table 1). The onset of primary aHUS occurs during adulthood in 41–58 % of patients.^{76–78}

Pathogenesis

The underlying pathogenesis is dysregulation of the complement AP, which is mainly restricted to cell surfaces. Three complement activation pathways - the classical, lectin, and alternative pathways - generate protease complexes, called C3 and C5 convertases, which cleave C3 and C5, respectively, ultimately leading to the formation of the membrane attack complex (C5b-9), which causes cell lysis.⁷⁹ Unlike the other pathways, the AP is initiated spontaneously in plasma by hydrolysis of C3 (Fig. 4). On host cells, complement activation is tightly controlled by fluid-phase and membrane regulators, while on foreign targets and injured cells that either lack membrane-bound regulators or cannot bind soluble regulators, the complement attack occurs selectively (Fig. 4). The main regulators are the plasma protein factor H (FH) and I (FI) and the membrane cofactor protein (MCP). FH and MCP act as co-factors of factor I in the proteolytic inactivation of C3b. Factor H regulates the AP of complement, both in the fluid phase and on cells, by accelerating the decay of C3bBb and by competing with FB to bind C3b.⁷⁹

About 50–60% of aHUS patients carry abnormalities affecting genes that code complement regulators (*CFH*, *MCP*, *CFI* and *THBD*) and components (*C3* and *CFB*) of the C3 convertase.^{6,76,78,80}

Pathogenic variants in *CFH* are the most frequent, accounting for 19–26% of childhood cases and 25–40% of adult patients.^{76–78} *CFH* abnormalities most

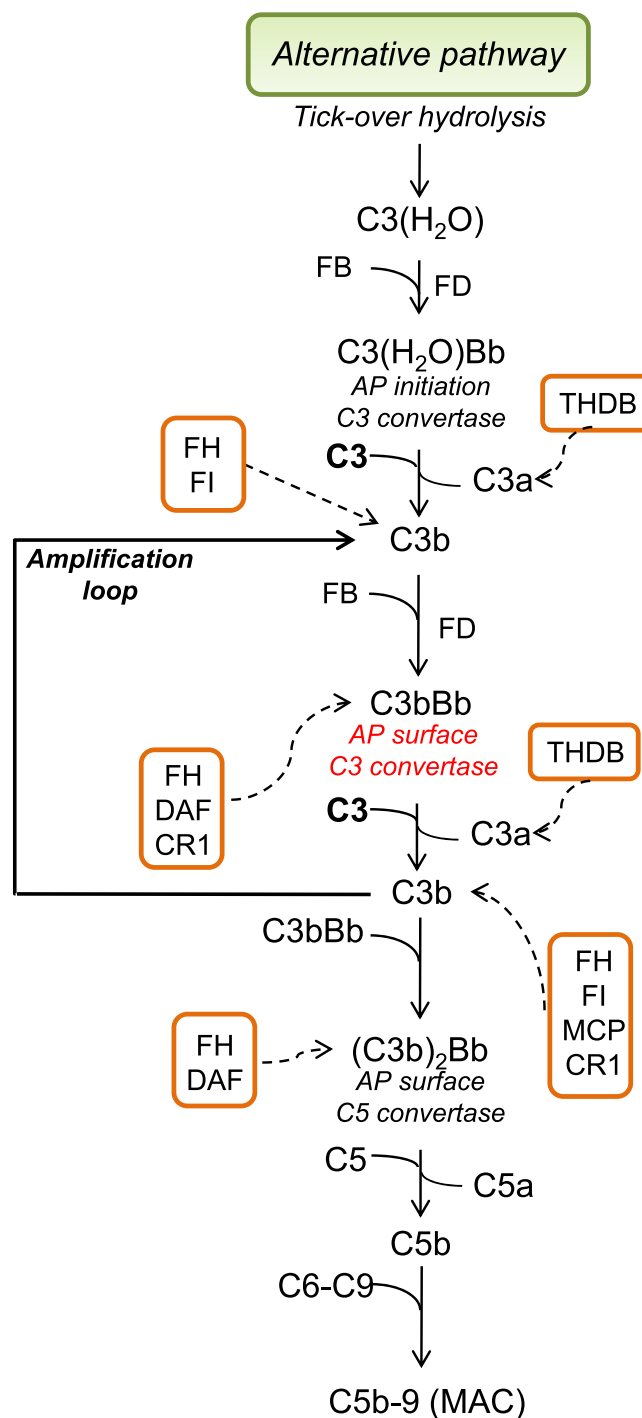


Figure 4. Schematic representation of the complement alternative pathway activation. The AP is continuously activated in plasma by low grade hydrolysis of C3 forming C3(H₂O). The latter binds to factor B (FB), which in turn is cleaved by factor D (FD) to form the AP fluid phase C3 convertase, C3(H₂O)Bb. The C3 convertase cleaves C3 into C3a and C3b. Once formed, C3b contributes to the formation of the AP surface C3 convertase that cleaves additional C3 molecules, resulting in an amplification loop. In addition, C3b contributes to the formation of the C5 convertases that cleave the component C5 leading to the formation of C5a and C5b. C5b initiates the late events that culminate in the formation of the membrane attack complex (MAC or C5b-9 complex). Self surfaces are protected from complement damage by protein regulators: FI factor I; DAF decay accelerating factor; MCP membrane cofactor protein; CR1 complement receptor 1; FH factor H; THBD thrombomodulin.

commonly reduce FH capacity to bind and regulate complement on endothelial cells and platelets.⁸¹ Genomic rearrangements involving *CFH* and *CFHRs* have been identified in 3-8% of patients and impair regulatory activity on cell surface.^{82,83} *CFI* pathogenic variants are more common in aHUS patients with adult onset (10-12%) than in pediatric cases (3-6%),⁷⁶⁻⁷⁸ and variant proteins are mostly not secreted or if secreted have functional deficiency.^{84,85}

Variants in *MCP* are associated with earlier onset than *CFH* or *CFI* variants (9-14% in children versus 4-8% adults).⁷⁶⁻⁷⁸ *MCP* mutants have reduced C3b-binding and cofactor activity and/or decreased surface expression.⁸⁴ Gain-of-function *CFB* variants are rare, accounting for 1-2% childhood and adulthood cases. *C3* variants are identified in 4-7% and 8-11% of cases with onset in childhood and adulthood, respectively,^{76-78,86} and are usually associated with low C3 levels. Most pathogenic variants reduce C3b binding to regulators. About 3% of patients carry variants in *THBD* encoding thrombomodulin, an anticoagulant protein expressed on the endothelial surface, which also modulates complement.⁸⁷ *THBD* mutants inactivate less efficiently either C3b⁸⁷ or the anaphylatoxins C3a and C5a.⁸⁸

Combined pathogenic variants are found in 1.3–5.6% of patients.⁸⁹

Complement gene haplotypes, mainly *CFH-H3* and *MCPgaac*, may also predispose to aHUS, and their concomitant presence in carriers of pathogenic variants increases aHUS penetrance.^{76,77,90}

Anti-FH autoantibodies account for 8-25% and 2-6% of cases with childhood and adult onset, respectively,^{76-78,91}. A higher prevalence is observed in non-Caucasians, reaching 24% of cases in children in Egypt, 30% in Korea and 56% in India.⁹²⁻⁹⁴ Similar to *CFH* genetic defects, anti-FH autoantibodies mostly target the C-terminus, thereby impairing complement regulation on cell surfaces. The development of autoantibodies in aHUS has a genetic predisposition, being strongly associated with the homozygous deletion of *CFHR1* and *CFHR3*.⁹⁵

In over 50% of patients with aHUS and carriers of genetic risk factors, disease development is preceded by a trigger, including infections, pregnancy and drugs.^{76,96} A gastrointestinal or respiratory viral prodrome was observed in 39% and 8% of children, respectively, but was uncommon in adults (15% and 1%).⁷⁷ Approximately 20% of aHUS in adults occur in association with pregnancy, usually at term or post-partum.⁹⁷ Approximately 50-60% of pregnancy-associated aHUS cases show complement genetic abnormalities⁹⁶ and increased *ex-vivo* serum-induced C5b-9 formation on endothelial cells.⁹⁸

Clinical features

Acute episodes manifest with severe hemolytic anemia, thrombocytopenia and kidney injury. Thrombocytopenia

is usually mild, normal platelet counts ($>150 \times 10^9/L$) have been reported in 15-20% of children and adults.⁷⁶ Renal involvement may be less severe in children than in adults: 59% of children vs. 81% of adults required dialysis at presentation.⁷⁶

Before the eculizumab era, 48-59% of children and 64-67% of adults progressed to kidney failure or died within 1-5 years.^{76,77,99} Fifty to 70% of patients with *CFH*, *CFI*, *C3*, *FB*, or *THBD* variants and 30-40% of those with anti-FH antibodies had a poor outcome.⁵ Progression to kidney failure occurred most rapidly in patients with *CFH* variants.⁷⁸ In patients with *MCP* variants, recurrences were frequent, but the long-term outcome was good with 80% of patients remaining dialysis-free.⁵ Eculizumab has radically changed the natural history of aHUS. The vast majority of aHUS patients respond to C5 inhibition (see Management section).

Extrarenal involvement occurs in 12-47% of children and 8-33% of adults.⁷⁶⁻⁷⁸ The most frequent extrarenal symptoms are neurological (16-24% of children and 8-25% of adults)^{77,78} (Table 2). Cardiac events (myocardial infarction, cardiomyopathy, heart failure and peripheral ischemic heart disease) have been reported in 3-10% of pediatric and adult patients^{100,101} (Table 2). Hypertensive emergency at presentation has been reported in 54% of adult patients.¹⁰² Gastrointestinal manifestations were observed in a higher proportion of children than adults (47% vs. 33%)⁷⁸ (Table 2). Pulmonary manifestations occur more commonly in adults than children (20% versus 12%).⁷⁸ Other findings, chiefly in children, include asymptomatic pancreatitis (~7%), hepatitis (6%), multi-organ failure (3%), pulmonary hemorrhage (2%), and pericarditis (1%).⁷⁷

Diagnosis

The diagnosis of aHUS is made by exclusion of concurrent diseases or STEC-HUS or Sp-HUS, or severe ADAMTS13 deficiency. In adults, clinical context and symptoms at presentation are more confounding than in children and can make it difficult to rule out other forms of TMA. For this reason, aHUS has to be suspected whatever the clinical context, and additional investigation is usually needed. The diagnosis is more straightforward in childhood, and kidney biopsy is rarely required [Fig. 3 (b and c); Fig. 5 (a-d)].

There is no gold standard laboratory assay that can enable a direct diagnosis. Complement analysis in cases of suspected aHUS should include serum levels of C3, C4, FH, FI and FB and plasma levels of Bb. However, circulating complement profile not always highlights complement dysregulation, which in aHUS is mainly restricted to the cell surface. Indeed, about 50% of patients have normal C3 levels, even during the acute phase, and normal C3 levels should not rule out a complement-related aHUS.¹⁰³

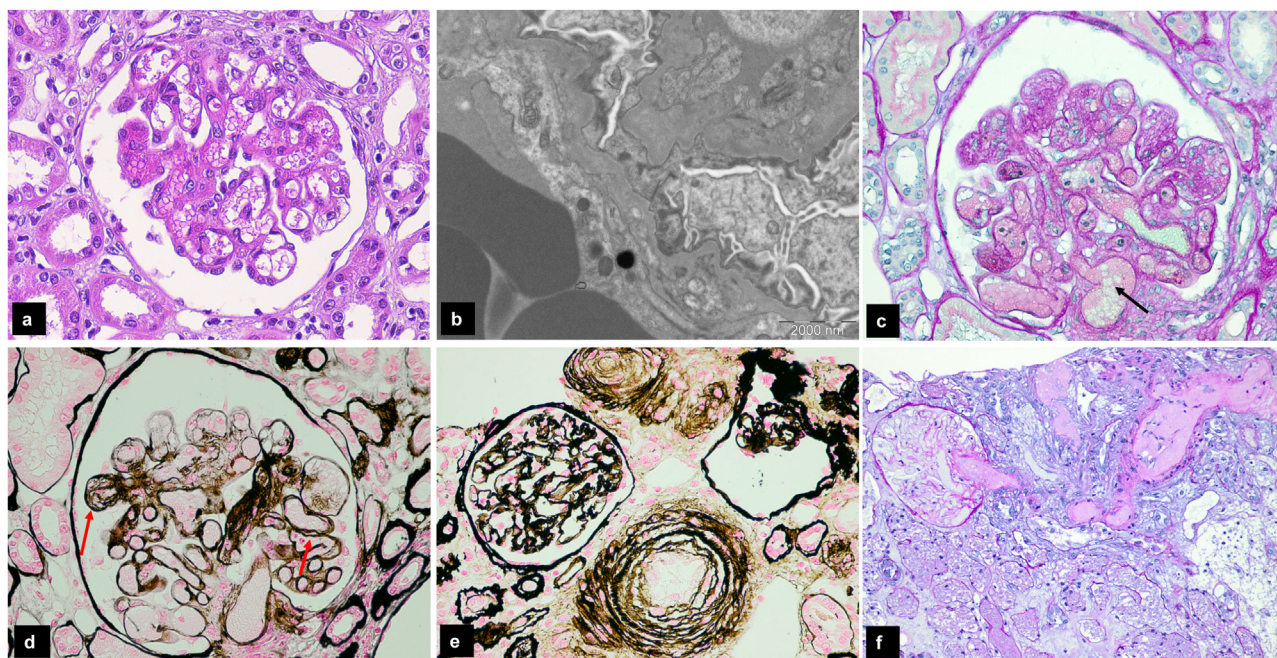


Figure 5. Spectrum of thrombotic microangiopathy (TMA) in adults with primary and secondary atypical hemolytic uremic syndrome (HUS). (a) A glomerulus with abundant erythrocytes and capillary lumina occluded with fibrin in an adult male with aHUS carrying a heterozygous *CFH* pathogenic variant (Hematoxylin and eosin, x 200 original magnification); (b) electron microscopy of the same patient revealed swollen endothelium and endothelial detachment; (c) A glomerulus showing endocapillary hypercellularity, fibrin deposits within glomerular capillary lumina and occlusion of the feeding arteriole (arrow) in an aHUS patient carrying a heterozygous *CFH* pathogenic variant (Periodic acid Schiff, x 200 original magnification); (d) the same glomerulus also showed extensive capillary wall reduplication (arrows) and fibrillary mesangium (Jones's silver methenamine 200x magnification); (e) Cortical renal parenchyma showing glomeruli with tuft shrinkage and Bowman's capsule thickening, and intimal arterial thickening in a patient with aHUS secondary to malignant hypertension (Jones's silver methenamine 100x magnification); (f) Low magnification of renal parenchyma showing diffuse tubular necrosis and thrombosis of a medium-sized interlobular arteriole and of its branches, including the glomerular tributary arteriole, in a patient with systemic lupus erythematosus with TMA secondary to antiphospholipid syndrome (Periodic acid Schiff, x 100 original magnification).

Screening for disease-associated genes and for anti-FH antibodies is recommended in the diagnosis of aHUS.¹⁰⁴ However, failure to find any pathogenic variant does not exclude a diagnosis of aHUS.⁸⁰ Although new sequencing techniques have reduced time and cost, a complete evaluation still takes up to several weeks and therefore cannot guide early treatment.

In search of a clinical tool to detect complement dysregulation at cell-surface in aHUS, an *ex-vivo* assay based on serum-induced C5b-9 formation on human microvascular endothelial cells (HMEC-1) has been developed, which is useful for rapid diagnosis of complement-mediated aHUS.¹⁰⁵

Management

Until 2011, plasma therapy, including infusion (PI) or TPE, was the only available therapy for aHUS. In adults presenting with a TMA of unclear etiology, TPE remains the first line treatment, due to the prevalence of TTP in adults. Consensus-based guidelines recommend that therapy (TPE, 1-2 plasma volumes/day; PI, 20-30 ml/kg/day) be started within 24 hours of diagnosis, and

continued until resolution of TMA (or a declaration of no-response). Patients with anti-FH autoantibodies benefit from TPE in association with immunosuppressive therapy.¹⁰⁴ However, long-term treatment may fail due to development of plasma resistance.

The anti-C5 antibody eculizumab has revolutionized treatment for aHUS and, where available, it is now used as a first-line therapy, provided that other causes of TMA are excluded.¹⁰⁶ Eculizumab treatment efficiently antagonizes terminal complement hyperactivation on cell surface in aHUS, as documented by complete normalization of serum-induced C5b-9 formation on HMEC-1.¹⁰⁵ In adults, the high efficacy and safety of eculizumab in inducing remission of acute episodes and sustaining long-term remission, both in native kidneys and in the kidney grafts, has been documented in three prospective trials.^{107,108} Good outcomes after eculizumab have been reported in aHUS triggered by pregnancy as well, and the drug was safe for the mother and the fetus.⁹⁷ A systematic review of 15 studies including 940 pediatric patients, showed that eculizumab improved renal and hematological parameters in most children, with a low (1.6%) mortality rate.¹⁰⁹

Meningococcal prophylaxis and vaccination are mandatory.

Duration of eculizumab therapy is unclear, a crucial issue because the drug is very expensive. A prospective multicenter study of eculizumab discontinuation showed that during follow-up, 31% of children and 19% of adults experienced aHUS relapse, and in multivariable analysis, the presence of a rare complement gene variant was associated with an increased risk of relapse.¹¹⁰ A new prospective study of eculizumab discontinuation, CUR-EiHUS, reported that, during follow-up, relapse occurred in four patients (3 adults and 1 child). All carried at least one complement pathogenic variant.¹¹¹ Therefore, clinicians should be cautious when discontinuing eculizumab in this patient population. Results from an ongoing prospective trial (SETaHUS, United Kingdom) will further guide practice.

Reliable biomarkers of early relapse are urgently needed. A rise in serum-induced C5b-9 deposition on HMEC-1 heralded aHUS relapses during eculizumab discontinuation.¹⁰⁵

Recently, an anti-C5 antibody engineered from eculizumab, ravulizumab, has been approved to treat aHUS. Qualitative review of four single-limb interventional trials on eculizumab and one on ravulizumab indicates complete TMA response in 60% and 54% patients, respectively, at 26 weeks.¹¹²

Kidney transplantation

Before the availability of anti-complement therapy, the outcome of kidney transplantation in aHUS was poor because of recurrence. Disease recurred in 60-80% of transplanted patients with variants in complement circulating proteins, and graft failure occurred in 80-90%.¹¹³ A study in adult aHUS kidney transplant recipients showed four times higher risk of recurrence for patients with *CFH* pathogenic variants, or *CFH/CFHR1* hybrids, or *CFB/C3* abnormalities, compared to patients without these genetic abnormalities.¹¹⁴

Transplant protocols that integrate eculizumab treatment have been an important advance. The KDIGO workgroup recommends the prophylactic use of eculizumab for kidney transplant patients who are at high risk of recurrence based on the genetic background.¹⁰⁴

A meta-analysis of 18 studies in 380 adult patients with aHUS who underwent transplantation, indicated low rates of recurrence (6.3%) and allograft loss (5.5%) after prophylactic eculizumab administration, while the risk of allograft loss in patients receiving eculizumab at the time of aHUS recurrence was 22.5%.¹¹⁵

Combined Liver-Kidney Transplant

In patients with *CFH* abnormalities, combined liver-kidney transplants have been carried out to correct the

genetic defect. Other complement proteins, including FI, FB, and C3 are produced in the liver and liver transplantation could correct the genetic abnormality in patients who carry pathogenic variants. Liver transplantation, in contrast to eculizumab therapy, cures aHUS definitively without the need for specific therapies other than standard immunosuppression to prevent graft rejection.

Non-complement-mediated aHUS

Rarely, aHUS is caused by homozygous or compound heterozygous variants in *DGKE* encoding diacylglycerol kinase epsilon (*DGKE*), mainly in very young children presenting with normal serum C3 levels and therapy-refractory disease.^{116,117} The condition is quite rare (Table 1). However, 2-25% of all cases below 2 years of age, and 50% of familial cases in this age group, are secondary to *DGKE* nephropathy. *DGKE* is an intracellular lipid kinase which phosphorylates diacylglycerol conjugated with arachidonic acid, to phosphatidic acid. Absence of *DGKE* leads to unbalance of pro- and anti-thrombotic factors that maintain stability of renal microvasculature, and alterations in vascular endothelial growth factor signaling that impact podocyte and endothelial function.¹¹⁶

Typically, patients present with clinical and histological features of aHUS before 1-2 years of age, often with hypertension, hematuria and heavy proteinuria, and sometimes with nephrotic syndrome.^{116,118-120} Six of 14 patients in an international study had developmental delay, autism or learning disability.¹¹⁹ Histologically, chronic thrombotic microangiopathy mimicking a membranoproliferative pattern, often with foot process effacement and without significant arteriolar changes, develops over 3-5 years [Fig. 3(d-e)].^{116,118} Patients usually do not respond to complement inhibitors, or immunosuppressive agents. Management is largely supportive and outcomes often unsatisfactory. Disease relapses are common, as is progression to ESKF, at median of two decades.^{116,119,120} Given the renal localization of the defect, the condition does not recur after kidney transplantation.

WT1 mutations are associated with rare cases of aHUS. *WT1* encodes a transcription factor that regulates the differentiation of progenitor cells, especially in the gonads and kidneys. Five patients with Denys-Drash syndrome, who initially presented with aHUS at 8-26 months of age¹²¹ were refractory to TPE and/or eculizumab, progressed to kidney failure, and did not show post-transplant recurrence. While the pathogenesis of aHUS remains unclear, reduced *WT1* expression might induce podocyte and/or endothelial injury.¹²¹

Metabolism-associated HUS

These uncommon entities, comprising less than 5% of all cases of TMA, may complicate cobalamin C (cblC)

disease, the most common disorder of cobalamin metabolism, and cobalamin G (cblG) disease, which are caused by homozygous or compound heterozygous mutations in *MMACHC* (encoding methylmalonic aciduria and homocystinuria type C) and *MTR* (encoding methionine synthase), respectively.^{122–124} Endothelial injury induced by hyperhomocysteinemia impairs nitric oxide-dependent inhibition of platelet aggregation, and induces an endothelial procoagulant state.

The diagnosis is suspected in young infants presenting with TMA associated with neurological features, normal serum complement C3 level and refractoriness to conventional therapy.

Screening tests include plasma levels of homocysteine, amino acids (methionine) and methylmalonic acid.

Mortality is high if untreated, but metabolic therapy with hydroxycobalamin is very effective.

Secondary aHUS

Secondary HUS occurs chiefly in adults (Fig. 5 (e-f)). The prevalence of genetic defects in secondary HUS is highly variable, ranging from almost 60% in cases associated with malignant hypertension to less than 10% in drug-induced HUS or HUS associated with autoimmune diseases.^{8,69,98,125,126}

Even in the absence of genetic abnormalities, the involvement of complement in the pathogenesis of secondary HUS cannot be ruled out as second hit. This possibility is supported by excessive complement deposition on HMEC-1 exposed to serum and plasma from patients, and by measurements of circulating complement parameters.^{105,127,128}

HUS and malignant hypertension

AP complement defects were identified in 35–65% of patients who develop HUS in the setting of malignant hypertension (HUS-MHT)⁸ (Fig. 5e). C5b-9 deposits were observed in renal biopsies of HUS-MHT patients¹²⁹ and serum induced massive C5b-9 formation on endothelial cells in 69–87% of cases, not evident in patients with MHT alone.^{105,126,127} Treatment involves lowering blood pressure. Good response to C5 inhibitor therapy has been reported.^{130,131}

HUS associated with autoimmune and systemic disease

Systemic lupus erythematosus (SLE), antiphospholipid syndrome (APS), and scleroderma may be associated with HUS (Fig. 5f). Complement genetic abnormalities in these patients are rare.⁶⁹ The first line of intervention is to treat the underlying autoimmune disease. In SLE, eculizumab improved renal and hematological outcomes in a few adult patients with persistent TMA.^{125,132} In APS syndrome, eculizumab has been successfully used

in adult patients refractory to treatment with steroids, anticoagulation, TPE, and rituximab.¹³² Good response to eculizumab strengthens the role of complement activation in scleroderma.⁸

Drug-induced HUS

Two main mechanisms of drug-induced HUS have been described: direct toxicity and immune mediated damage. Drugs such as interferon β , chemotherapeutic agents, and immunosuppressive agents may cause direct endothelial toxicity. On the other hand, quinine, one of the most common drugs causing TMA, can raise autoantibodies that target platelets and endothelium. Remission is common upon early drug discontinuation. Complement gene variants are rare,⁶⁹ and the response after drug withdrawal was similar in patients treated with plasma, steroids or eculizumab.^{8,69}

Post-transplant HUS

HUS can occur *de novo* in patients who receive kidney or other solid organ transplants [Fig. 3 (f)]. HUS may occur in 5–10% and 1% of patients who receive cyclosporine and tacrolimus, respectively. The inhibitors of mammalian target of rapamycin (mTORi) either alone or in combination with cyclosporine may increase the risk of *de novo* PT-TMA.¹³³ Complement gene variants have been identified in approximately 33% of patients.¹¹⁴ Resolution can be observed when the drug is discontinued or tapered.⁸ Eculizumab may be effective, although further studies are needed to establish which patients would benefit.¹³³ HUS occurs in 10–20% of patients receiving allogeneic hematopoietic stem cells and is associated with a high mortality rate.¹³⁴ Good outcomes with eculizumab have been reported in a few cases.

Thrombotic thrombocytopenic purpura (TTP)

Epidemiology

TTP is more common in adults (Table 1). iTTP accounts for 95% of adult cases.^{135–137} Both in adults and children, iTTP is more common in females than in males (ratio 2:1). cTTP is very rare and accounts 10% of all cases of TTP¹³⁶ (Table 1).

Pathogenesis and etiology

TTP is caused by a severe deficiency of ADAMTS13, the protease responsible for cleaving VWF.^{7,136} VWF, released from endothelial cells as ultra-large multimers (UL-VWF), is cleaved by ADAMTS13 into smaller fragments, which are important in hemostasis. In patients with TTP, ADAMTS13 deficiency leads to the accumulation of highly adhesive UL-VWF. Under high shear

stress, found in arteriolar and capillary circulation, platelet attachment to these UL-VWF causes unregulated platelet adhesion, aggregation and ultimately the formation of microvascular thrombi in almost all organs, causing ischemia, and thrombocytopenia secondary to platelet consumption. Compared with HUS, thrombi are more extensively distributed, probably reflecting the more systemic nature of the disease.^{2,138}

Acquired immune-mediated TTP (iTTP), accounts for almost 90% of cases, and is characterized by specific anti-ADAMTS13 autoantibodies, mostly IgG, which tend to disappear during remission.¹³⁹ Inhibitory antibodies prevent the proteolytic activity of ADAMTS13, while non-inhibitory antibodies bind to the enzyme, accelerating its clearance from plasma, or alter its synthesis or secretion. The majority of iTTP cases are primary or idiopathic. iTTP secondary to malignancies, HIV infection, drugs, or various autoimmune conditions, tend to have less severe ADAMTS13 deficiency.¹⁴⁰ An exception is iTTP secondary to ticlopidine, where 80-90% of patients have severe ADAMTS13 deficiency and inhibitory antibodies.¹⁴¹

Congenital TTP (cTTP, Upshaw-Shulman disease) accounts for 10% of cases. It is an autosomal recessive condition caused by compound heterozygous (80%) or homozygous (20%) *ADAMTS13* mutations,¹³⁹ resulting in deficient protease secretion or severely impaired proteolytic activity. The type and location of *ADAMTS13* variants might affect the age of onset and disease severity.¹³⁶ Variants resulting in very low ADAMTS13 activity (<1%) are associated with childhood onset, relapsing disease and AKI.¹⁴²

Clinical course

The clinical manifestations of TTP include severe thrombocytopenia, microangiopathic hemolytic anemia, and a variable degree of ischemic organ damage, particularly affecting the brain, heart and kidneys. If left untreated, mortality exceeds 90%.¹³⁶ Over 80% of patients have neurological manifestations, which range from headache, mild confusion to stroke, seizures, or coma.^{139,143} Compared with STEC-HUS or aHUS, acute TTP in adults usually presents with more severe thrombocytopenia and milder AKI.¹⁴⁴ Gastrointestinal ischemia and cardiac involvement are present in approximately 35% and 25% of acute patients, respectively¹³⁹ (Table 2).

Children with iTTP have similar rates of AKI, association with autoimmunity and relapse as adults with iTTP.¹⁴⁵ Fever was present at onset in one-third cases, extrarenal involvement was reported in 64% cases.¹⁴⁵ cTTP may present in the neonatal period with severe jaundice, anemia and thrombocytopenia, however the majority of patients present within 5 years of age, without any sex predilection. However, a high proportion of cases is diagnosed in young women, usually in the

context of pregnancy. The first and subsequent episodes of TTP may be spontaneous or triggered by infections, vaccination, pregnancy and surgical procedures. Almost 50% of the cases are severe and chronic, requiring monthly prophylactic plasma therapy.

Diagnosis

The initial diagnosis of TTP is frequently clinical, since ADAMTS13 activity testing is limited to expert and reference laboratories.¹⁴⁶ Composite scores, such as the PLASMIC, Bentley and French scores, have been developed in order to have prognostic probability of identifying TTP patients and to avoid delays in initiation of treatment.^{147,148} However, severe ADAMTS13 deficiency, defined by an activity level <10%, is mandatory to confirm the diagnosis of TTP. The gold standard assay for ADAMTS13 activity is the fluorescence resonance energy transfer (FRETs) method; other commonly used methods are the ELISA and collagen binding assay (CBA).¹⁴⁶ Once TTP is confirmed, further diagnostic tests include ADAMTS13 autoantibody assays to identify iTTP. The diagnosis of cTTP is confirmed by the presence of *ADAMTS13* mutations.

Management

The primary treatment for iTTP is TPE, which supplies suitable levels of ADAMTS13 while removing anti-ADAMTS13 autoantibodies. Typically, treatment consists of a daily 1-2 plasma volume exchanges until the platelet count is stably normal ($>150 \times 10^9/L$).¹⁴⁶

The International Society of Thrombosis and Haemostasis (ISTH) recommends in association with TPE the use of corticosteroids to inhibit the synthesis of anti-ADAMTS13 autoantibodies,^{149,150} and suggests the addition of rituximab.¹⁵¹ Rituximab was associated with clinical remission, the disappearance of anti-ADAMTS13 antibodies, and an increase in ADAMTS13 activity to >10%. The standard dosing is 375 mg/m² weekly for a total of four doses.^{146,152,153} Retreatment with rituximab is considered when ADAMTS13 decreases and inhibitors reappear, to prevent a relapse.¹⁵⁴ The ISTH suggests considering rituximab prophylaxis to prevent relapses in patients with iTTP in clinical remission with low ADAMTS13 activity.¹⁵¹

Two randomized clinical trials found that caplacizumab, an anti-VWF humanized single-variable-domain antibody, along with TPE and immunosuppression therapy, significantly reduced time to platelet count normalization as compared with placebo. By inhibiting interaction between VWF multimers and platelets, caplacizumab reduced microvascular thrombosis, but as the production of anti-ADAMTS13 autoantibodies was not affected, concomitant immunosuppression and TPE was required.¹⁵⁵

In cTTP, ADAMTS13 is constitutively lacking and can be replaced by plasmatherapy. During acute episodes, the patients can be successfully treated with plasma infusions (10-15 ml/kg/day) until clinical response is achieved. Regular prophylactic plasma infusions every 2-3 weeks should be considered in patients with a recurring cTTP phenotype.^{137,156} Solvent/detergent treated (to inactivate lipid-enveloped viruses) pooled plasma (OCTAPLAS®) may be used instead of fresh frozen plasma, if available.

In the near future rhADAMTS13, successfully tested in a Phase 1/2 study in cTTP patients (NCT03393975), will greatly facilitate regular prophylactic treatment. A Phase 3 trial is currently underway (NCT04683003).

Outcomes

Long-term follow-up of TTP survivors reveals many potential complications and morbidity in addition to the risk of relapse.^{137,157} More than 50% of patients who survived their initial acute event and were older than 40 years, experienced a major morbidity during follow-up: stroke, kidney injury requiring dialysis, and/or severe cardiac injury.¹⁵⁷

Complement activation as a common pathogenic effector

A large body of evidence has shown that overactivation of the complement system, with its prothrombotic and proinflammatory effects, is not restricted to aHUS but is a common pathogenic effector in other forms of TMA, as well.¹⁵⁸ Complement system activation on the endothelium has been observed in STEC-HUS.¹⁸ Several *in vivo* and *in vitro* findings have provided evidence that UL-VWF accumulation on the endothelial cell surface in the context of ADAMTS13 deficiency stimulates the activation of the complement AP until the terminal pathway, with the formation of C5a and C5b-9.¹⁵⁹ UL-VWF multimers may act as platform for the assembly of AP C3 convertase,¹⁵⁹ and, in addition, impair the regulation of AP. In turn, the terminal complement products lead to the loss of the anti-thrombotic THBD and the increase in VWF multimers on endothelial cells, thus generating a positive amplification loop that results in microvascular thrombosis.^{158,159} Evidence of the potential role of the complement activation in these diseases could pave the way for new treatment opportunities for the most severe cases of STEC-HUS and TTP.

CONCLUSIONS

The spectrum of TMA is broad and encompasses a wide range of conditions in both pediatric and adult populations. Typically, in children, the most prevalent forms of TMAs are STEC-HUS, pneumococcal HUS, antibody-

mediated aHUS, DGKE-associated HUS, and congenital TTP, while among adults, aHUS secondary forms and immune-mediated TTP are more common. The wide spectrum of etiologies underscores the importance of a timely and precise diagnosis to guide targeted therapy. Although the past decade has witnessed great advances in the pathophysiology of TMA syndromes, there is still much work to be done in order to untangle the intricate relationships and distinctions observed among the various subclasses.

DECLARATION OF COMPETING INTEREST

RD has received honoraria from Alexion Pharmaceuticals for contributing to a publishing project.

MN has received honoraria from Alexion Pharmaceuticals for giving lectures and for participating in advisory boards, and she has received research grants from Omeros, Gemini, Novartis and BioCryst Pharmaceuticals. GR has consultancy agreements with AbbVie, Alexion Pharmaceuticals, Novartis Pharma and BioCryst Pharmaceuticals.

ACKNOWLEDGEMENTS

The Authors acknowledge Dr Paola Rizzo and Dr Camillo Carrara to having supplied the histological images of adult patients. The authors are also grateful to Kerstin Mierke for English language editing.

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